A PROJECT SYNOPSIS ON

DEALDISH

A Sustainable Marketplace for Surplus Hotel Food

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in partial fulfillment of the requirements for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

DEPARTMENT OF COMPUTER APPLICATIONS

NEHRU COLLEGE OF ENGINEERING AND RESEARCH CENTRE (NCERC)

Approved by AICTE, Affiliated to APJ Abdul Kalam Technological University

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JULY 2026

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Background of Food Wastage & Surplus Management

Food wastage has become one of the major global challenges of the 21st century, contributing significantly to environmental pollution, economic losses, and resource depletion. Every day, hotels and restaurants prepare a substantial variety and quantity of food based on estimated customer demand. However, due to unpredictable customer turnout, sudden booking cancellations, and seasonal variations, a considerable quantity of freshly prepared, high-quality food remains unsold at the end of daily operational hours. In traditional food management frameworks, this surplus food is discarded, leading to direct commercial losses and contributing to municipal waste disposal concerns.

1.2 The Need for a Centralized Marketplace

To mitigate this structural waste, 'DealDish' is conceived as a real-time, centralized digital marketplace that bridges the communication gap between food businesses and local consumers. By providing a secure and interactive platform, hotels can catalog their surplus meals at discounted prices shortly before closing hours. Consumers, specifically students, local residents, and budget-conscious individuals, can browse active listings in their vicinity, reserve items online, and collect them within the allotted timeframe. This ecosystem enables businesses to recover a portion of their raw material investments while offering affordable food alternatives, encouraging responsible consumption and waste reduction.

2. PROBLEM STATEMENT

Currently, hotels and restaurants in Kerala operate in isolation. There is no dedicated channel for identifying and distributing edible surplus food. Due to strict food safety guidelines, unsold items cannot be stored for next-day operations, leading to massive disposal patterns. The primary pain points identified in this manual ecosystem include:

- **Food Wastage and Environmental Degradation:** Large quantities of edible, fresh surplus food are discarded daily into public waste disposal systems, worsening public hygiene concerns, filling municipal landfills, and contributing significantly to greenhouse gas emissions during decomposition.

- **Financial Losses for Hospitality Partners:** Hotels and restaurants absorb the complete cost of production, ingredients, labor, and utility expenses for all unsold food. The lack of a secondary recovery channel prevents them from retrieving even minimal operational capital, resulting in direct financial deficits.
- **Lack of Consumer Awareness:** While local students, low-income groups, and budget-oriented individuals are seeking affordable meals, they remain unaware of high-quality surplus meals available at nearby hotels due to the absence of an informational interface.
- **Absence of a Centralized Platform:** There is no real-time dashboard or reservation ledger that connects hotels and consumers. This prevents businesses from listing surplus inventory dynamically, and prevents customers from reserving and picking up food securely before it expires.

3. OBJECTIVES

The primary goal of 'DealDish' is to centralize Kerala's surplus food management and redistribution system. The project is focused on achieving the following clear objectives:

1. Centralization of Surplus Listings: Establish a unified web application where verified hotels can instantly upload their surplus food inventory along with pickup windows and discounted prices.

2. Real-Time Quantity Tracking: Implement an algorithmic transaction module that monitors active listings and updates quantities dynamically as reservations are placed.

3. Role-Based Access Environments: Design tailored dashboard portals for three primary user roles: Platform Administrators, Hotel Managers, and Customers, ensuring data privacy and workflow integrity.

4. Verification Gates for Safety: Integrate a document compliance workflow requiring hotels to upload valid FSSAI food safety licenses for admin approval before they can list food items.

5. Modern Sustainability Design: Develop a responsive and interactive Single Page Application (SPA) utilizing a modern, green-themed design system, encouraging sustainable consumer behavior.

4. SCOPE OF THE PROJECT

The boundaries and operational workflow of the DealDish marketplace are designed around three primary user categories:

- **Hotel Managers:** Register business profiles, submit FSSAI licenses, list surplus food items (specifying name, category, price, discount, quantity), and view reserved/completed pickups.
- **Customers:** Register/login, view verified nearby food listings, filter by category/distance, place food reservations, track order histories, and collect orders within specified pickup windows.
- **Platform Administrators:** Audit and approve newly registered hotels by verifying submitted credentials, monitor active listings, manage system users, and extract transaction analytics.

The project prototype operates as a fully client-side Single Page Application (SPA), utilizing an emulated relational database on top of HTML5 localStorage for offline compatibility and ease of demonstration. Future horizons include deploying a complete Node.js/Express REST API backend, implementing MongoDB for cloud storage, and integrating online payment escrow gateways.

5. PROPOSED SYSTEM

5.1 System Architecture Overview

The proposed DealDish application is built as a clean, modular Single Page Application (SPA) using TypeScript and modern web standards. To bypass complete page refreshes and simulate mobile-app fluidity, the client-side bootstrap initialises a custom regular-expression-based router inside router.ts. This router intercepts browser anchors and dynamically binds views. The emulated data-access layer in db.ts manages schema constraints across local database models (Users, Hotels, Food Items, Reservations) serialized directly into HTML5 localStorage. Changes in collection states trigger custom DOM events, allowing headers, analytics counters, and catalog lists to update immediately.

5.2 Reservation and Expiry Logic

To handle the highly perishable nature of surplus hotel food and prevent system conflicts, the platform enforces two core validation algorithms within its transaction layer:

1. Surplus Food Quantity Verification Algorithm: Surplus portions are treated as finite inventory assets. When a customer submits a reservation request for quantity 'Q' of a food item, the transaction engine triggers a schema lock. The engine checks the current inventory state. If the requested quantity satisfies the inequality Q <= (QuantityTotal - ReservedQuantity), the reservation is successfully approved, and the reserved quantity is incremented. Otherwise, the transaction is rejected, and an 'Insufficient Stock' exception is returned to the user interface. This validation occurs both at request creation and state confirmation to resolve race conditions from concurrent users.

2. Pickup Window and Expiry Check Algorithm: To ensure public health safety standards, surplus food listings are strictly time-bound. Each listing is assigned a pickup window defined by [PickupStart, PickupEnd] (typically aligning with one hour prior to hotel closing). A background daemon or route-middleware evaluates the system time against the PickupEnd parameter. If CurrentTime > PickupEnd, the listing's status is modified to 'expired'. Any pending, uncollected reservations associated with this listing are automatically cancelled, and notification events are dispatched to release temporary ledger allocations. This ensures that customers can never collect food that has exceeded safety thresholds.

5.3 Database Relational Schema

The following database schemas define the emulation layer of DealDish. They are mapped with unique IDs acting as Primary Keys (PK) and references acting as Foreign Keys (FK):

Table 5.3.1: Users Schema (users)

Field Name	Data Type	Key	Description
id	String	PK	Unique identifier for the user (u-xxx)
name	String	-	Full name of the user or authority
email	String	-	Login email address (unique value)
passwordHash	String	-	SHA-256 representation of password
role	Enum	-	admin | hotel | customer
phone	String	-	Mobile contact number
district	String	-	District of operations in Kerala
isVerified	Boolean	-	Flag indicating account status clearance

Table 5.3.2: Hotels Schema (hotels)

Field Name	Data Type	Key	Description
id	String	PK	Unique identifier for the hotel profile (h-xxx)
ownerId	String	FK	References users.id (role: hotel)
hotelName	String	-	Registered business name of the hotel
address	String	-	Physical location address of the establishment
fssaiNumber	String	-	Food safety license registration number
closingTime	String	-	Standard operational closing time (HH:MM)
isApproved	Boolean	-	Status set by admin after profile audit
updatedAt	Date	-	Timestamp of last profile details update

Table 5.3.3: Food Items Schema (foodItems)

Field Name	Data Type	Key	Description
id	String	PK	Unique identifier for the food listing (f-xxx)
hotelId	String	FK	References hotels.id (role: hotel)
itemName	String	-	Commercial name of the surplus meal
category	Enum	-	veg | non-veg | bakery | beverage
originalPrice	Number	-	Standard menu price in local currency
discountedPrice	Number	-	Listing price for discounted surplus sale
quantityAvailable	Number	-	Number of food portions available
pickupStart	String	-	Start time for surplus collection (HH:MM)
pickupEnd	String	-	End time for surplus collection (HH:MM)
status	Enum	-	active | reserved | expired

Table 5.3.4: Reservations Schema (reservations)

Field Name	Data Type	Key	Description
id	String	PK	Unique identifier for the reservation (r-xxx)

reservation details

customerId	String	FK	References users.id (role: customer)
foodItemId	String	FK	References foodItems.id
quantity	Number	-	Number of portions reserved by customer
totalPrice	Number	-	Total price of reservation (Qty * discountedPrice)
reservationTime	Date	-	Timestamp of reservation creation
pickupStatus	Enum	-	pending | completed | cancelled | expired
status	Enum	-	pending | accepted | rejected | confirmed

6. MODULES OF THE PROJECT

DealDish is divided into six core modules, each handling a distinct segment of functionality:

1. Auth & Role Module (auth.ts): Manages user signup, role selection (Hotel, Customer, Admin), login, and secure sessions using client-side SHA-256 hashing and sessionStorage.

2. Hotel Management Module (hotels.ts): Allows hotel administrators to upload and manage business profiles, FSSAI registration details, operational hours, and view dashboard analytics.

3. Surplus Listing Module (foodItems.ts): Enables hotel kitchen staff to list surplus meals, specify quantity, original price, discounted price, and pickup window schedules.

4. Reservation & Search Module (reservations.ts): Allows customers to browse available listings, search by category or location, view discounts, and reserve surplus food portions.

5. Admin Moderation Module (dashboard.ts): Provides administrative oversight, listing pending hotel registrations and FSSAI licenses to verify them before they go public.

6. Routing & UI Engine (router.ts, layout.ts): Implements a custom regex router to intercept page requests and render views dynamically. Features a responsive green-themed design.

7. SOFTWARE AND HARDWARE REQUIREMENTS

The development and execution environments for DealDish are detailed below:

Table 7.1: Software Requirements

Component	Specification
Operating System	Windows 10/11, Linux (Ubuntu), or macOS
IDE / Editors	Visual Studio Code (v1.85+)
Runtime & Package Manager	Node.js (v18.x+) & NPM
Programming Language	TypeScript (v5.x) & ES6+ JavaScript
Bundler / Server	Vite (v5.x) development compiler
Target Browsers	Chrome (v115+), Firefox, Edge, or Safari
Database	HTML5 LocalStorage (Emulated relational database)

Table 7.2: Hardware Requirements

Component	Specification
Processor	Intel Core i3 or AMD Ryzen 3 (2.4 GHz minimum)
Memory (RAM)	8 GB DDR4 RAM
Storage System	256 GB SSD (with 10 GB free space)
Display Screen	Standard 15.6-inch screen (1366x768 resolution)

8. EXPECTED OUTCOME

- **No Over-Allocation Conflicts:** The quantity check algorithm will completely prevent over-allocation of food items, ensuring real-time listing accuracy.
- **Surplus Sales Recovery:** Hotel operators can track surplus sales and reclaim marginal food production costs, preventing immediate financial loss.
- **Safe Food Distribution:** Strict verification gates will block unlicensed food establishments, raising the safety standards of listed items.
- **Environmental Waste Mitigation:** Providing a structured marketplace will reduce the volume of organic food waste thrown into municipal trash bins.

9. CONCLUSION

DealDish successfully digitizes and optimizes one of the most resource-intensive and wasteful sectors of the hospitality industry: surplus food management. By bridging the information gap between hotels and consumers, it adds transparency, cost-efficiency, and ecological responsibility to daily operations. The Single Page Application (SPA) architecture utilizing TypeScript offers a high-performance experience, while the localized relational database emulates concurrency and reservation management. Overall, DealDish acts as a blueprint for blending social responsibility with modern software engineering, ensuring a sustainable food ecosystem.

10. REFERENCES

Below are the primary resources, specifications, and academic guides referenced during the development:

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